

JÚLIO SILVA

Backend Engineer

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🌐 [jmbcs.github.io/portfolio](#)



Backend engineer with six years building production systems — REST APIs, OTA channel integrations, and distributed event pipelines. Focused on API design, data consistency, and services that stay reliable under load. Python and Django.

Work Experience

Backend Engineer GuestReady / RentalReady

📅 Oct 2024 – Present 📍 Remote

- Built and maintained live channel integrations with **Booking.com**, **Airbnb**, and **Vrbo** — real-time sync of availability, rates, and reservations for thousands of short-term rental properties.
- Designed and maintained clean **REST API** contracts between internal services and external OTA channels.
- Delivered promotions features enabling dynamic discount campaigns across OTA channels.
- Built a **Redis** hash-based deduplication layer that prevents reprocessing of stale events in the channel pipeline.
- Built asynchronous and scheduled processing with **RabbitMQ** and **Celery** — including recurring jobs such as activating promotions when a property goes live.
- Ran zero-downtime bulk operations on large **PostgreSQL** tables in production.
- Profiled and optimized database queries to keep the channel pipeline performant under load.
- Implemented **New Relic** monitoring alarms for proactive observability; triaged production incidents with **Sentry** and **Datadog**.
- Worked across a large **Django** PMS codebase, investigating and resolving production bugs reported by users and support.
- Maintained engineering quality through **code review** on a distributed 40+ engineer team; used AI tooling (**Claude Code**, **Cursor**) to ship faster.
- Mentored junior developers, sharing backend best practices through code review and pairing.
- Delivered frontend work alongside backend, including a full revamp of the Amenities page.

Software Engineer Wavecom

📅 Jan 2021 – Oct 2024 📍 Aveiro, Portugal (Hybrid)

- Built microservices in **Python** and **Go**, using **gRPC** and **Protocol Buffers** for typed inter-service communication.
- Designed **REST APIs** over SQL and NoSQL stores; containerized services with **Docker** and **Docker Compose**.
- Architected a real-time monitoring platform for **HPC** clusters with **Grafana** and **VictoriaMetrics**; built **ETL pipelines** using **Telegraf** and **Pandas**.
- Automated build and deployment workflows with **Makefiles** and **Bash**; evaluated technologies through PoCs.

Education

MSc Electronics & Telecommunications University of Aveiro

📅 2014 – 2020 📍 Aveiro, Portugal

Thesis: Recovery and Identification of Moments 19/20

- Developed a **Python** image retrieval system associating images to moments described in natural-language text.

Tech Skills

Programming

Python SQL JavaScript React
HTML

Frameworks

Django FastAPI SQLAlchemy Pandas

APIs & Messaging

REST gRPC RabbitMQ Celery
Protocol Buffers

Databases

PostgreSQL Redis VictoriaMetrics

Infrastructure

Docker Linux AWS Qovery

Observability

Grafana Sentry Datadog New Relic
Telegraf

Tooling

Git Bash Make YAML

AI Tooling

Claude Code Cursor

Languages

Portuguese — Native English — Fluent

Relevant Links

BigHPC — [bighpc.wavecom.pt](#)
Conference Talk — [youtube.com](#)
Thesis — [ria.ua.pt](#)
Thesis Paper — [ceur-ws.org](#)